

MovieLOD

Linked movie data application · Semantic Web

From movie data to a knowledge graph

Ontology · RDF · Linked Open Data · SPARQL

Full 24–slide presentation

Team / members / class: [complete before submission]

30

real films

42

named classes

1,727

external links

Problem and five assignment requirements

Who did what in which film? Where did the data come from? How can we query it?

ID	Requirement	Deliverable / evidence
R1	Define an ontology	OWL/Turtle, classes, properties and restrictions
R2	Collect domain data	APIs, raw responses, URLs, timestamps and hashes
R3	Publish four-star data	RDF, HTTP IRIs, open license and publication
R4	Link to reach five stars	sameAs links, matching method and public data
R5	Provide SPARQL interfaces	Web application, endpoint and terminal

Required: report ≤15 pages · slides · 3–5-minute video. Running examples: Inception and Nolan.

Architecture and data flow

Asserted data, schema and inferred types are stored separately

01

COLLECT

Wikidata / DBpedia
collect.py → data/raw/

02

BUILD RDF

build.py → movies.ttl
+ ontology/movie.ttl

03

VALIDATE

validate.py / pytest
sources, fields and
queries

04

CLASSIFY

reason.py →
inferred_classes.ttl

APPLICATION: Web / Flask + RDFLib / Comunica / Terminal
Endpoint: movies.ttl. --reasoned: adds schema + inferred_classes.ttl.

Current dataset: interpreting the numbers

A selected sample of 30 films · evidence/statistics.json

30

films

851

people

1,010

contributions

45

companies

672

award entities

76/76

verified source hashes

19,339 data triples · 518 schema triples · 75 genres · 11 countries · 15 languages

R1 · Ontology: 42 classes in six groups

Direct reuse of dbo:Film, dbo:Person and dbo:Country

11 classes	CreativeWork / Film
8 classes	Agent / Person / Organization
6 classes	Contribution / ContributionRole
8 classes	Genre
5 classes	Award
4 classes	Country / Language / SourceSnapshot / Dataset

3 DBpedia classes + 39 ex: classes
14 classes receive types; 42 classes explicitly declared.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P00

Ontology IRI và phiên bản

P00_ontology_header.png

1. Mở ontology/Movie_Ontology.owl.
2. Chọn Active Ontology / Ontology Header.
3. Giữ IRI ontology và versionInfo 2.0.0 trong khung hình.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

R1 · Creative works and film classes

Class definitions and additional instance types · (I) marks additional classification

CreativeWork

dbo:Film

FeatureFilm · 29 films
 AnimatedFilm · 1 film
 DocumentaryFilm · 0 films
 ActionFilm (I)
 ComedyFilm (I)
 DramaFilm (I)
 ScienceFictionFilm (I)
 MultiGenreFilm (I)
 AwardWinningFilm (I)

ẢNH PROTÉGÉ CẦN BỔ SUNG · P01

Cây lớp phim

P01_film_hierarchy.png

1. Entities → Classes → mở CreativeWork và Film.
2. Hiện các lớp con FeatureFilm, AnimatedFilm và nhóm phim.
3. Chọn Film; giữ IRI DBpedia và Class Description.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

(I): additional types. FeatureFilm/AnimatedFilm follow P31 mapping; DocumentaryFilm has no sample instances.

R1 · People, organizations and roles

A person may be both an Actor and a Filmmaker

Agent

dbo:Person · 851 people

Actor (I) · 769

Filmmaker (I) · 89

AwardWinner (I) · 290

Organization

ProductionCompany · 45

FilmStudio · 6 by counting; DL: 0

ẢNH PROTÉGÉ CẦN BỔ SUNG · P02

Cây người và tổ chức

P02_agent_hierarchy.png

1. Entities → Classes → mở Agent.
2. Mở Person và Organization → ProductionCompany.
3. Chọn Person hoặc Filmmaker; giữ cây lớp và Description.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Person and Organization are different kinds of agent. Actor and Filmmaker may overlap.

R1 · Genres, awards and provenance

Each branch supports a specific group of questions

Genre

FictionGenre

ActionGenre

ComedyGenre

DramaGenre

ScienceFictionGenre

NonFictionGenre

DocumentaryGenre

Award

FilmAward

ActingAward

DirectingAward

WritingAward

ẢNH PROTÉGÉ CẦN BỔ SUNG · P03

Cây Genre và Award

P03_genre_award_hierarchy.png

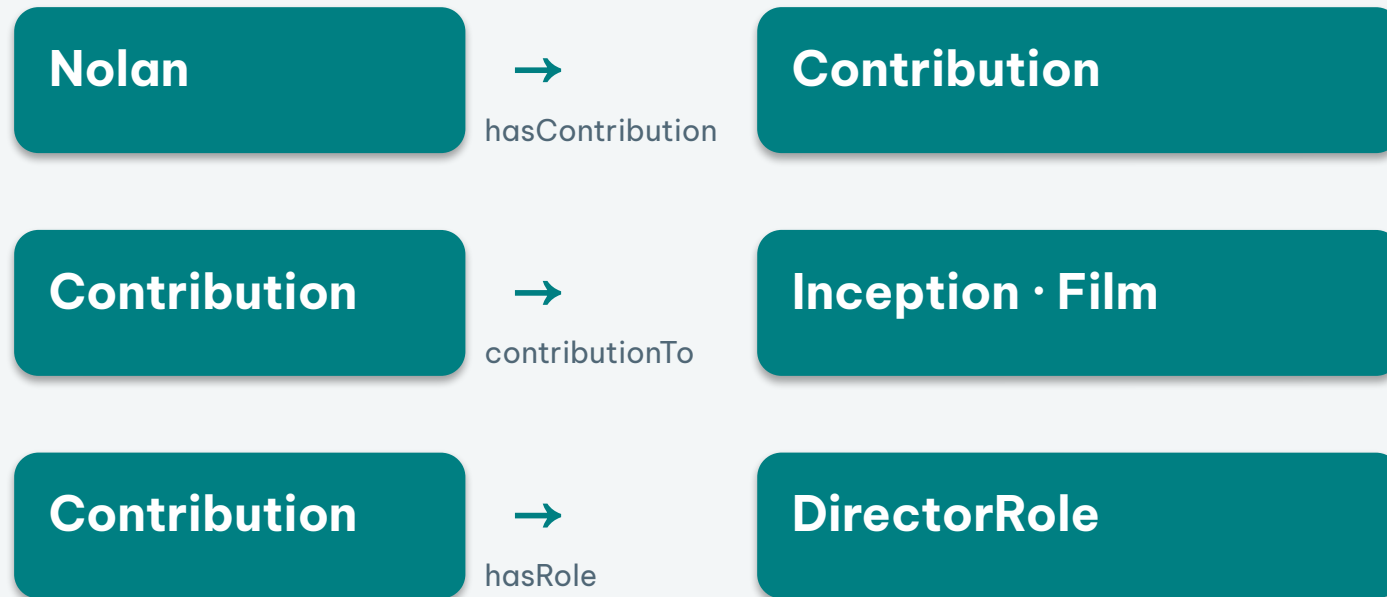
1. Thu gọn Film/Agent để ảnh dễ đọc.
2. Mở Genre → FictionGenre / NonFictionGenre và Award.
3. Hiện các lớp thể loại và nhóm giải; giữ Class Description.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Supporting classes: dbo:Country · Language · SourceSnapshot · Dataset. Group mapping uses label keywords.

R1 · Contribution: who did what in a film?

One record connects exactly one person, one film and one role



Each role has its own record

Director / Actor / Writer / Producer. Nolan has 3 roles in Inception.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P04

Ràng buộc của Contribution

P04_contribution_restrictions.png

1. Entities → Classes → chọn Contribution.
2. Trong Class Description, mở đủ SubClass Of.
3. Hiện contributionBy / contributionTo / hasRole: exactly 1.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

R1 · Properties, domain/range and inverses

23 object properties · selected central relationships

Property	Domain → range	Inverse
contributionBy	Contribution → Person	hasContribution
contributionTo	Contribution → Film	contributionOf
hasRole	Contribution → Role	roleOf
dbo:director	Film → Person	directed
hasGenre	Film → Genre	genreOf
hasAward	Film or Person → Award	awardOf

ẢNH PROTÉGÉ CẦN BỔ SUNG · P05

Domain, range, inverse, functional

P05_object_property.png

- 1. Entities → Object properties → chọn contributionBy.
- 2. Hiện Domain=Contribution, Range=Person.
- 3. Giữ inverse hasContribution và ô Functional được chọn.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

contributionBy / contributionTo / hasRole: functional. Domain/range can infer types; inverse reverses a relation.

R1 • How OWL expresses meaning

Ontology constraints and application validation serve different purposes

Class definitions

equivalentClass + AND / OR / SOME
A Person with filmmaking contributions → Filmmaker.

Cardinality and functional

Contribution: exactly one person / film / role.
Functional: at most one value under OWL semantics.

Inverse and disjointness

director ↔ directed; hasRole ↔ roleOf.
Actor and Filmmaker are not disjoint.

Open-world assumption

Missing information ≠ negation.
Python validates required fields; OWL defines meaning.

R1 · Fourteen classes with additional types

None of these 14 types is asserted in the original data graph

Class	Condition in plain language	Count	Method
ActingContribution	Contribution with ActorRole	855	OWL RL
DirectingContribution	Contribution with DirectorRole	31	OWL RL
WritingContribution	Contribution with WriterRole	51	OWL RL
ProducingContribution	Contribution with ProducerRole	73	OWL RL
Actor	Person with acting contributions	769	OWL RL
Filmmaker	Person with filmmaking contributions	89	OWL RL
AwardWinner	Person who receives an award	290	OWL RL
ActionFilm	Film with ActionGenre	12	OWL RL
ComedyFilm	Film with ComedyGenre	4	OWL RL
DramaFilm	Film with DramaGenre	25	OWL RL
ScienceFictionFilm	Film with ScienceFictionGenre	6	OWL RL
AwardWinningFilm	Film that receives an award	26	OWL RL
MultiGenreFilm	Film with at least 2 Genre IRIs	30	IRI counting
FilmStudio	Company with at least 3 Film IRIs	6	IRI counting

12 classes: OWL RL, with matching HermiT/Pellet counts. Last 2: COUNT DISTINCT; DL instances: 0.

R1 · Inference example: Christopher Nolan

DirectorRole → DirectingContribution → Filmmaker

FACTS

Nolan a dbo:Person
c hasRole DirectorRole



RULE 1 · hasValue

c a DirectingContribution



RULE 2 · SOME + OR

Nolan a Filmmaker

ẢNH PROTÉGÉ CẦN BỔ SUNG · P06

Định nghĩa Filmmaker

P06_filmmaker_equivalent_class.png

1. Entities → Classes → chọn Filmmaker.
2. Mở Class Description → Equivalent To.
3. Hiện Person AND các nhánh SOME Directing/Writing/ProducingContribution.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Nolan: Person (asserted) → Person + Filmmaker + AwardWinner (with inferred types).

R1 · Cardinality: thresholds of two and three

Distinguish OWL definitions from counting distinct RDF terms

MultiGenreFilm

Film + min 2 hasGenre.Genre

Application: 30/30 films; DL: 0 inferred.

FilmStudio

ProductionCompany + min 3 productionOf.Film

Application: 6 companies; DL: 0 inferred.

Distinct IRIs ≠ provably different OWL individuals.

Local namespace filter + COUNT(DISTINCT IRI) avoids counting sameAs aliases as separate values.

ẢNH PROTÉGÉ CẦN BỔ SUNG · P07

Định nghĩa FilmStudio

P07_filmstudio_cardinality.png

1. Entities → Classes → chọn FilmStudio.
2. Mở Equivalent To; hiện min 3 productionOf Film.
3. Có thể chụp thêm MultiGenreFilm: min 2 hasGenre Genre.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

R2 · Real collection and traceable sources

76/76 raw responses present · 76/76 SHA-256 hashes match

Resolve the right entity

Wikidata: exact Wikipedia sitelink.
DBpedia: matched root must be typed Film.

Collect relevant claims

P57 / P161 / P58 / P162: people and roles.
P166: awards; P272: companies; year/runtime/genre.

Trace and reproduce

URL + retrieval time + HTTP status + SHA-256.
Cached collect → build → validate rerun.

R2 · Actual Wikidata response and provenance

Read from the saved source file; SHA-256 verified. This is a source viewer, not a Protégé screenshot.

Response metadata

```
{
  "provider": "Wikidata",
  "url": "https://www.wikidata.org/w/api.php?action=wbgetentities&format=json&languages=en&sites=enwiki&titles=Inception%7CInterstellar+%28film%29%7CThe+Dark+Knight%7CMemento+%28film%29%7CDunkirk+%282017+film%29%7CThe+Prestige%7CBatman+Begins%7CTenet%7CThe+Godfather%7CThe+Godfather+Part+II&props=labels%7Cdescriptions%7Csitelinks%7Cclaims",
  "retrieved_at": "2026-10-06T08:43:21.455429+00:00",
  "sha256": "56d8b57c83cb1ad3dc2d2e0cf47035801fede896cc0dca296a6e8ff1e0c246e1"
,
  "path": "data/raw/1db5b6a369cc85140c7afe5e.json",
  "http_status": 200
}
```

Selected raw fields (references omitted)

```
{
  "id": "Q25188",
  "labels": {
    "en": {
      "language": "en",
      "value": "Inception"
    }
  },
  "P57_director": [
    {
      "mainsnak": {
        "snaktype": "value",
        "property": "P57",
        "hash": "c163ceb80831ed3f753fb3b90ec8c31d153eb0bf",
        "datavalue": {
          "value": {
            "entity-type": "item",
            "numeric-id": 25191,
            "id": "Q25191"
          },
          "type": "wikibase-entityid"
        },
        "datatype": "wikibase-item"
      }
    }
  ]
}
```

Right: actual Wikidata metadata and response excerpt. Hashes verify integrity, not real-world truth.

R3 · RDF and typed literal values

19,339 data triples · 518 schema triples · Turtle / JSON-LD / OWL

```
res:film-Q25188 a dbo:Film ;
ex:title "Inception" ;
ex:releaseYear 2010 ;
ex:runtimeMinutes "148.0"^^xsd:decimal ;
dbo:director res:person-Q25191 .
```

ẢNH PROTÉGÉ CẦN BỔ SUNG · P08

Datatype của thời lượng

P08_runtime_datatype.png

- 1. Entities → Data properties → chọn runtimeMinutes.
- 2. Hiện Domain=Film và Range=xsd:decimal.
- 3. Có thể chụp thêm releaseYear có range xsd:integer.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Data property	xsd: type
title	string
releaseYear	integer
runtimeMinutes	decimal
sourceUrl	anyURI
retrievedAt	dateTime
sha256	string

R3–R4 · From RDF to Linked Open Data

Star levels are cumulative · publication and links are verified

1 star	Public data + open license	CC BY–SA 4.0
2 stars	Structured data	Films, people, roles and attributes
3 stars	Open formats	CSV / Turtle / JSON–LD
4 stars	HTTP IRIs + RDF + lookup	HTML, JSON–LD and RDF descriptions
5 stars	Links to external datasets	1,699 Wikidata + 28 DBpedia

sameAs: shared identity. sourceSnapshot: provenance. All four checked public RDF graphs match local data.

R3 · Looking up Inception's IRI

One identifier offers readable and machine-readable descriptions

← MovieLOD

Inception

<https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/resource/film-Q25188>

Download RDF (Turtle)

Property	Value
director	Christopher Nolan
producer	Emma Thomas
producer	Christopher Nolan
starring	Dileep Rao
starring	Michael Caine
starring	Elliot Page
starring	Joseph Gordon-Levitt
starring	Tohoru Masamune
starring	Claire Geare
starring	Cillian Murphy
starring	Tom Hardy
starring	Ken Watanabe
starring	Pete Postlethwaite
starring	Taylor Geare
starring	Talulah Riley
starring	Tom Berenger

ẢNH PROTÉGÉ CẦN BỔ SUNG · P09

Cá thể Inception

P09_inception_individual.png

- 1. Mở ontology/Movie_Knowledge_Graph.owl.
- 2. Individuals → chọn film-Q25188 (Inception).
- 3. Hiện Film, title, releaseYear=2010, runtimeMinutes=148 và director.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

Left: real application screenshot. Right: pending Protégé evidence for Inception in the main Knowledge Graph.

R5 · Querying Inception on the Web

SELECT → title, year, runtimeMinutes and director

```
SELECT ?title ?year ?runtimeMinutes ?director
WHERE {
  ?film a dbo:Film ; ex:title "Inception" ;
  ex:title ?title ; dbo:director ?person .
  ?person rdfs:label ?director .
  OPTIONAL { ?film ex:releaseYear ?year }
  OPTIONAL { ?film ex:runtimeMinutes ?runtimeMinutes }
}
```

2010 · 148 minutes · Nolan

SELECT chooses columns; WHERE matches; OPTIONAL preserves rows.

Use the PREFIX declarations in queries/02_inception.rq. Actual result: 2010 · 148 minutes · Christopher Nolan.

MovieLOD

QueriesFilmsOntologyDataset

Download RDF ↗

SEMANTIC WEB / MOVIES

Ask questions about movie data.

301,72742

filmsexternal linksclasses

Explore films with source records, stable identifiers and links to Wikidata and DBpedia.

SPARQLQuery language

Sample queries

Inception: year, runtime and director

PREFIX ex: <https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?title ?year ?runtimeMinutes ?director WHERE {
 ?film a dbo:Film ; ex:title "Inception" ; ex:title ?title ;
 dbo:director ?person .
 ?person rdfs:label ?director .
 OPTIONAL { ?film ex:releaseYear ?year }
 OPTIONAL { ?film ex:runtimeMinutes ?runtimeMinutes }
}

Run queryCtrl + JCopy

SELECT: get a table · ASK: test a condition · CONSTRUCT: create an RDF graph.

Results

Download results

1 result · 0.01 s

title	year	runtimeMinutes	director
Inception	2010	148.0	Christopher Nolan

Querying the application SPARQL endpoint.

Reading an RDF statement: Inception → director → Christopher Nolan. ex: is the project vocabulary; dbo: is the DBpedia vocabulary; variables start with ?.

COLLECTED DATA

Find a film

Film title or director

R5 · One person, multiple roles and questions

The screenshot filters Nolan: Director, Writer and Producer in Inception

MovieLOD

QueriesFilmsOntologyDataset

Download RDF ↗

SEMANTIC WEB / MOVIES

Ask questions about movie data.

Explore films with source records, stable identifiers and links to Wikidata and DBpedia.

30

1,727

42

filmsexternal linksclasses

SPARQL

Query language

Sample queries

Inception: year, runtime and director

PREFIX ex: <https://movie-lod-semantic-web.laminhtrung2001.chatgpt.site/ontology#>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
SELECT ?personName ?roleName WHERE {
 ?film ex:title "Inception" ; ex:contributionOf ?contribution .
 ?contribution ex:contributionBy ?person ; ex:hasRole ?role .
 ?person rdfs:label ?personName . ?role rdfs:label ?roleName .
 FILTER(STR(?personName) = "Christopher Nolan")
} ORDER BY ?roleName ?personName

Run query ⌘↵Copy

SELECT: get a table · ASK: test a condition · CONSTRUCT: create an RDF graph.

Results

Download results

3 results · 0.03 s

personName	roleName
Christopher Nolan	Director
Christopher Nolan	Producer
Christopher Nolan	Writer

Querying the application SPARQL endpoint.

Reading an RDF statement:

Inception → director → Christopher Nolan. ex: is the project vocabulary; dbo: is the DBpedia vocabulary; variables start with ?.

COLLECTED DATA

Films in the dataset

Find a film

Film title or director

25 records

All Inception contributions; the image shows only Nolan.

8 / 4 / 7

Nolan films / Inception companies / Godfather awards.

14 Web samples / 24 files

Hierarchy requires schema; inferred types require --reasoned.

Data checked: 08 Oct 2026 • Sources: project code and evidence/

20 / 24

R5 · Endpoint, terminal and inferred types

Same Nolan query: one asserted type → three with saved inference

```
curl -X POST http://127.0.0.1:8000/sparql \
-H "Content-Type: application/sparql-query" \
--data-binary @queries/02_inception.rq

.venv/bin/python src/query.py \
queries/24_nolan_asserted_types_only.rq
.venv/bin/python src/query.py \
queries/24_nolan_asserted_types_only.rq --reasoned
```

Mode	Nolan types returned
Asserted graph	dbo:Person
With inference	Person + Filmmaker + AwardWinner

ẢNH PROTÉGÉ CẦN BỔ SUNG · P10

Kiểu khai báo của Nolan

P10_nolan_asserted_types.png

- 1. Mở Movie_Knowledge_Graph.owl, chọn person-Q25191 (Nolan).
- 2. Hiện Types trong chế độ asserted / trước suy luận.
- 3. Giữ dbo:Person và các quan hệ đóng góp/giải trong ảnh.

Thay khung bằng ảnh thật; chi tiết thêm trong Speaker Notes.

--reasoned loads schema + saved types. Endpoint: asserted graph by default. SELECT/ASK: JSON; graph queries: Turtle.

Validating data, application and publication

Execution evidence is stored in evidence/

76/76

Matching source hashes

24

Query files executed

14 + 9

Tests + browser checks

8 / 4

Public HTTP 200 / graph match

```
MovieLOD · Actual verification results

14 passed, 2 warnings in 96.97s (0:01:36)

Hermit 1.3.8.1099 · Movie_Knowledge_Graph.owl
Consistent: Yes · unsatisfiable named classes: 0

{
  "ActingContribution": 855,
  "DirectingContribution": 31,
  "WritingContribution": 51,
  "ProducingContribution": 73,
  "Actor": 769,
  "Filmmaker": 89,
  "AwardWinner": 290,
  "ActionFilm": 12,
  "ComedyFilm": 4,
  "DramaFilm": 25,
  "ScienceFictionFilm": 6,
  "AwardWinningFilm": 26,
  "MultiGenreFilm": 0,
  "FilmStudio": 0
}

Data checks: True
Source hashes: True · 76/76
SPARQL files executed: 24

Sources: evidence/tests.txt, hermit_run.json, validation.json
This is an execution-log viewer, not the Protégé interface.
```

14 tests / 9 browser checks. Hermit/Pellet verify DL separately; the image shows actual execution logs.

Assessment and limitations

Evidence for 5/5 requirements · 10/10 is a proposed self-score

Req.	Evidence	Score
R1	Ontology + OWL + tests	2/2
R2	76 sources + hashes + reproducibility	2/2
R3	Public RDF matches local data	2/2
R4	1,727 published identity links	2/2
R5	Web / endpoint / terminal	2/2

Total 10/10 · proposed self-score, not an official grade

Data limitations

Selected 30-film sample; label-based groups.
Hashes ≠ truth; budget/streaming data absent.

Reasoning limitations

12 OWL RL classes; 2 IRI-counting rules.
HermiT/Pellet: consistent; threshold classes: 0.

Semantic Web value

Shared IRIs, vocabulary, links and provenance.
SQL can also answer many of these questions.

Conclusion and pending Protégé evidence

24 slides · five-page report · four-minute-fifty-second demo

Image	Slide	Evidence to capture
P00	5	Ontology IRI and version
P01	6	Film class hierarchy
P02	7	People and organizations
P03	8	Genre and award hierarchies
P04	9	Contribution restrictions
P05	10	Domain, range, inverse and functional
P06	13	Filmmaker definition
P07	14	FilmStudio cardinality
P08	16	Runtime datatype
P09	18	Inception individual
P10	21	Nolan asserted types

Before submission

1. Complete team / member details.
2. Insert 11 genuine screenshots.
3. Save them in evidence/protege/ for automatic replacement.
4. Rehearse with the speaker script and review the video.

Screenshot slots are placeholders, not captured evidence. Full English narration is in Speaker Notes.